Fusion Finance Voice AI POC

Pre-collection Day Minus 1 Confirmation Calls

Try It Now

🔗 **Live App:** https://voice-pilvo-production.up.railway.app

Enter your phone number, select a language, and click "Initiate Call" to receive a test call.

What We Built

An automated voice calling system for Fusion Finance's pre-collection workflow. The system calls borrowers one day before their scheduled RO (Relationship Officer) visit to:

1. **Confirm availability** — Borrower presses 1 to confirm
2. **Capture reschedule reason** — Borrower presses 2, then selects from 6 decline reasons
3. **Handle unclear inputs** — Retries twice, then marks as "Not Responsive"

All interactions appear in a **live dashboard** with real-time transcripts and an AI analyst chat panel (simulated for POC).

Call Flow

```
📞 System initiates call
    ↓
🔊 Greeting plays: "[RO Name] will visit tomorrow. Press 1 to confirm, 2 to
reschedule"
    ↓
┌──────────────────────────────────────────────────────────┐
│  Borrower presses 1     →  "Thank you for confirming"  │
│                              ✅ OUTCOME: AVAILABLE      │
├──────────────────────────────────────────────────────────┤
│  Borrower presses 2     →  Decline reason menu plays   │
│                              (1=Travel, 2=Health, etc.) │
│      ↓                                                  │
│  Borrower selects reason →  "We will reschedule"       │
│                              🔄 OUTCOME: DECLINED       │
```

```
│                                 📝 REASON: [captured]      │
├──────────────────────────────────────────────────────────────┤
│  Invalid input (3,4,5..)→  "Press 1 or 2" (retry 1/2)  │
│  Invalid again          →  Retry 2/2                   │
│  Invalid 3rd time       →  Hang up                     │
│                                 ❌ OUTCOME: DECLINED       │
│                                 📝 REASON: NOT_RESPONSIVE  │
└──────────────────────────────────────────────────────────────┘
```

Supported Languages

Language	RO Name Used
Hindi	Amit ji
Telugu	Srinivas garu
Tamil	Suresh sir
Kannada	Shetty sir
Marathi	Patil saheb
English	Amit sir

Each language has 5 pre-recorded WAV audio files with natural, human voices.

Decline Reason Codes

Key	Reason
1	Travel / Out of town / Market visit
2	Health issue / Hospital
3	Financial stress / Cash flow
4	Work / Office commitment
5	Family event / Emergency

Key	Reason
6	Crop / Agriculture work

Architecture

Components

1. **Plivo** — Cloud telephony platform
    - Handles outbound calls to Indian mobile numbers
    - Captures DTMF (keypad presses)
    - Plays WAV audio files
2. **FastAPI Server** (Railway hosted)
    - Orchestrates call flow via Plivo XML webhooks
    - Serves audio files
    - Manages call state machine
    - Sends real-time updates via SSE
3. **Smaartbrand Dashboard**
    - Live transcript with DTMF events
    - Call initiation controls
    - AI analyst chat (simulated intelligence)
    - Language and phone number selection

Tech Stack

- Python 3.11 + FastAPI
- Plivo Voice API
- Server-Sent Events (SSE) for real-time updates
- Pre-recorded WAV audio (not TTS)
- Railway for hosting

Dashboard Features

Live Transcript Panel

- Real-time call events as they happen
- Color-coded by actor (System, Agent, Borrower)
- DTMF inputs highlighted with badges
- Timestamps for every event

AI Chat Analyst (Demo)

Simulated responses showcasing intelligence layer capabilities:

- "Why is Warangal high risk?" → Root cause analysis
- "Compare all clusters" → Cluster comparison table
- "How much can we save?" → ROI analysis (72x multiplier)
- "Show me the trend" → Weekly pattern insights

Call Controls

- Phone number input with country code
- Language dropdown (6 options)
- Initiate Call button
- Real-time status updates

Key Design Decisions

1. **WAV over TTS** — Pre-recorded audio in regional languages sounds more natural than text-to-speech, especially for Hindi, Tamil, Telugu, Kannada, and Marathi.
2. **DTMF over ASR** — Keypad input is 100% accurate vs speech recognition errors. Critical for capturing decline reasons.
3. **Retry Logic** — 2 retries before marking as "Not Responsive" balances patience with efficiency.
4. **State Machine** — Clean state transitions (INITIATED → GREETING → WAIT_CONFIRM → WAIT_REASON → COMPLETED) ensure predictable behavior.
5. **SSE for Real-time** — Server-Sent Events provide live updates without polling overhead.

Deployment

- **Platform:** Railway (auto-deploys from GitHub)
- **Repo:** github.com/giridharid/voice-pilvo
- **Version:** 1.1.3

Environment Variables (Railway)

```
PLIVO_AUTH_ID=MAOWI5YJMWNZQTYJQ0OS
PLIVO_AUTH_TOKEN=<secret>
PLIVO_PHONE_NUMBER=+918031320387
```

Next Steps (Post-POC)

1. **Sarvam.ai Integration** — Voice AI with ASR/TTS for conversational flows
2. **Campaign Management** — Batch calling with CSV upload
3. **Analytics Dashboard** — Real call data, not simulated
4. **CRM Integration** — Push outcomes to Fusion's loan management system
5. **Compliance** — DND checks, call timing windows, recording consent

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Value Proposition

For Fusion Finance:

- Automates 80%+ of confirmation calls
- Frees RO time for productive visits (not phone tag)
- Structured decline reasons enable portfolio-level insights
- Real-time visibility into call outcomes

Acquink's Role:

- Voice AI handles the call
- **Intelligence layer extracts signals from call data**
- Pattern detection across clusters, decline reasons, time trends
- Actionable insights for collection strategy

Questions?

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